Unit 4 :: Special Theory of Relativity

PHY0400104: Classical Mechanics

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Syllabus

Unit 4: Special theory of relativity

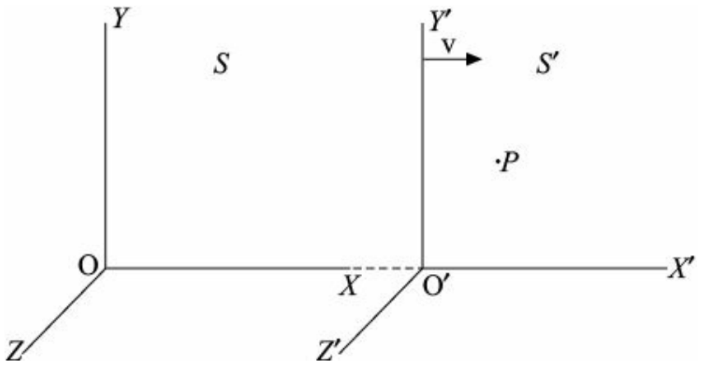
Inadequacy of Galilean transformation; postulates of special relativity; Lorentz transformation; simultaneity and order of events; length contraction and time dilation; relativistic addition of velocities; variation of mass with velocity and mass-energy equivalence. Lorentz transformation as a rotation in spacetime; relation between proper time and coordinate time; relativistic kinematics:energy-momentum relation.

1 Galilean transformation

An *inertial frame* is one in which the law of inertia holds good.

- An ideal inertial frame is a coordinate frame of reference which is fixed in space with respect to the fixed stars.
- A set of coordinate axes attached to the Earth may be regarded as an *almost* inertial frame of reference.

An accelerating frame of reference is known as a *non-inertial frame*.



Consider two inertial frames S and S' with coordinate axes xyz and $x'y'z'$ attached to them, with respective origins O and O' respectively. Let the frame S' be moving with respect to S with a uniform velocity v along the xx' axes as shown in the figure above. The origins of the two frames are assumed to coincide when $t = t' = 0$.

Let an event be taking place at P whose coordinate with respect to S be (x, y, z, t), and with respect to S' be (x', y', z', t'). Then these coordinates are related by the following *Galilean transformation equations*:

$$\boxed{\begin{aligned} x' &= x - vt \\ y' &= y \\ z' &= z \\ t' &= t \end{aligned}} \tag{1}$$

With the assumption that $\dfrac{\mathrm{d}}{\mathrm{d}t} = \dfrac{\mathrm{d}}{\mathrm{d}t'}$, we differentiate equation (1) with respect to time to obtain the *Galilean velocity transformations*:

$$\begin{aligned} \frac{\mathrm{d}x'}{\mathrm{d}t'} &= \frac{\mathrm{d}}{\mathrm{d}t}(x - vt) \\ \frac{\mathrm{d}y'}{\mathrm{d}t'} &= \frac{\mathrm{d}y}{\mathrm{d}t} \\ \frac{\mathrm{d}z'}{\mathrm{d}t'} &= \frac{\mathrm{d}z}{\mathrm{d}t} \end{aligned} \implies \boxed{\begin{aligned} u'_x &= u_x - v \\ u'_y &= u_y \\ u'_z &= u_z \end{aligned}} \tag{2}$$

Again we differentiate equation (2) with respect to time to obtain the *Galilean acceleration transformations*:

$$\begin{aligned}\frac{d^2x'}{dt'^2} &= \frac{d^2x}{dt^2} \\ \frac{d^2y'}{dt'^2} &= \frac{d^2y}{dt^2} \\ \frac{d^2z'}{dt'^2} &= \frac{d^2z}{dt^2}\end{aligned} \implies \boxed{\begin{aligned}a'_x &= a_x \\ a'_y &= a_y \\ a'_z &= a_z\end{aligned}} \tag{3}$$

Because the directions of the unit vectors are parallel in both frames, we have $\hat{\mathbf{x}}' = \hat{\mathbf{x}}$, $\hat{\mathbf{y}}' = \hat{\mathbf{y}}$ and $\hat{\mathbf{z}}' = \hat{\mathbf{z}}$. Then from equation (3), we find that

$$\begin{aligned} a'_x\hat{\mathbf{x}}' + a'_y\hat{\mathbf{y}}' + a'_z\hat{\mathbf{z}}' &= a_x\hat{\mathbf{x}} + a_y\hat{\mathbf{y}} + a_z\hat{\mathbf{z}} \\ \implies \mathbf{a}' &= \mathbf{a} \end{aligned} \tag{4}$$

i.e. the acceleration of a particle remains invariant under a Galilean transformation.

Multiplying equation (4) by the mass m of the particle, we have

$$\begin{aligned} m\mathbf{a}' &= m\mathbf{a} \\ \implies \mathbf{F}' &= \mathbf{F} \end{aligned} \tag{5}$$

Equation (5) implies that the force on a body remains invariant across inertial frames of reference.

Principle of Galilean-Newtonian relativity: No inertial frame is special and all inertial frames are equivalent.

1.1 The inadequacy of Galilean transformation

As per Maxwell's electromagnetic theory, an electromagnetic (EM) wave propagates through free space with a constant speed of $c \approx 3 \times 10^8$ m/s.

Then the wavefront of an EM wave emitted by a point source at O would be spherical with a radius at time t to be ct. Hence the spherical wavefront surface is given in the frame S to be

$$\begin{aligned} x^2 + y^2 + z^2 &= (ct)^2 \\ \implies x^2 + y^2 + z^2 - c^2t^2 &= 0 \end{aligned} \tag{6}$$

For the Galilean-Newtonian principle to hold, we must have that the above equation is invariant between S and S', and so we must have

$$x'^2 + y'^2 + z'^2 - c^2t'^2 = 0 \tag{7}$$

Substituting equations (1) in equation (7), we get

$$(x - vt)^2 + y^2 + z^2 - c^2t^2 = 0 \tag{8}$$

But equation (8) is not identical to equation (6). This inconsistency seems to suggest that there must be some special reference frame with respect to which the speed of light is 3×10^8 m/s.

In order to support this suggestion, it was assumed that light and other EM waves are mechanical waves, which thereby require a medium to propagate. This medium was given the name *ether*, was assumed to have the following properties:

i. Transparent and massless

ii. Permeates all space

iii. Rigid enough to support transverse motion of EM waves

The ether hypothesis led to the following two alternatives:

1. *The stationary ether hypothesis*: where the ether is at rest with respect to the bodies moving through is; the speed of light is always c in the ether frame.

2. *The ether drag hypothesis*: where the ether is dragged along with the bodies that move through it.

A number of experiments were performed to verify the ether hypothesis and all had failed. The most direct experiment was the Michelson-Morley experiment.

2 The postulates of special relativity

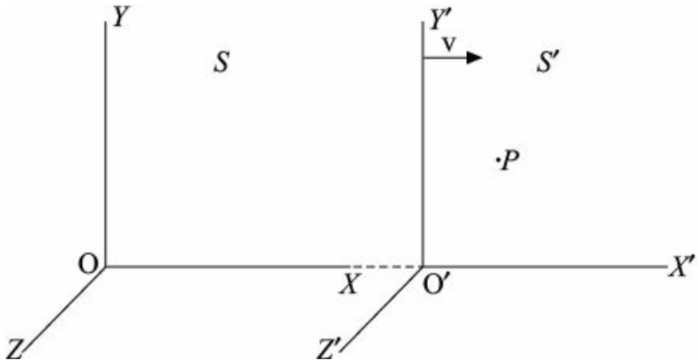
Due to the failure to detect the ether medium, Albert Einstein dropped the concept of ether, along with the assumption of an absolute rest frame.

Einstein revised the classical ideas of space and time in the form of the following two *postulates of special relativity*:

Postulate 1: The principle of equivalence
The laws of physics have the same form in all inertial frames of reference.

Postulate 2: The constancy of the speed of light
The speed of light in free space is always a constant c and is independent of the speed of the source, the observer, or the relative motion of the inertial systems.

3 Lorentz transformation



Consider two inertial frames S and S' with coordinate axes xyz and $x'y'z'$ attached to them, with respective origins O and O' respectively. Let the frame S' be moving with respect to S with a uniform velocity v along the xx' axes as shown in the figure above. The origins of the two frames are assumed to coincide when $t = t' = 0$.

Because the velocity of S' with respect to S is along the x-axis, we have

$$y' = y \quad \text{and} \quad z' = z$$

According to postulate 1, if the motion of a particle is uniformly rectilinear (i.e. in straight line) in frame S, then it must be uniformly rectilinear in frame S' as well. Hence the transformation relating x and x' must be linear, thereby written as

$$x' = \gamma(x - vt) \tag{1}$$

where γ is a dimensionless constant. The inverse transformation is then

$$x = \gamma(x' + vt') \tag{2}$$

Substituting for x' in equation (2) using equation (1), we have

$$\begin{aligned} x &= \gamma\big[\gamma(x - vt) + vt'\big] \\ &= \gamma^2 x - \gamma^2 vt + \gamma vt' \\ \implies \gamma vt' &= \gamma^2 vt + x - \gamma^2 x \\ &= \gamma^2 vt + (1-\gamma^2)x \\ \implies t' &= \gamma t + \frac{(1-\gamma^2)x}{\gamma v} \end{aligned}$$

Then we can write the tentative transformation equations as

$$\boxed{\begin{aligned} &x' = \gamma(x - vt) \\ &y' = y \\ &z' = z \\ &t' = \gamma t + \frac{(1-\gamma^2)x}{\gamma v} \end{aligned}} \tag{3}$$

Let us now consider an EM wave emitted by a point source located at the position O at time $t=0$. The spherical wavefront expands with a speed c. Hence the equation of the wavefront in the frame S is

$$x^2+y^2+z^2=c^2t^2 \tag{4}$$

The corresponding equation in the frame S' is

$$x'^2+y'^2+z'^2=c^2t'^2 \tag{5}$$

Now, using equations (3) in equation (5), we have

$$\begin{aligned}
\left[\gamma(x-vt)\right]^2+y^2+z^2 &= c^2\left[\gamma t+\frac{(1-\gamma^2)x}{\gamma v}\right]^2 \\
\Longrightarrow \gamma^2(x-vt)^2+y^2+z^2 &= c^2\left[\gamma^2t^2+\frac{(1-\gamma^2)^2x^2}{\gamma^2v^2}+2\gamma t\frac{(1-\gamma^2)x}{\gamma v}\right] \\
\Longrightarrow \gamma^2x^2+\gamma^2v^2t^2-2\gamma^2xvt+y^2+z^2 &= \gamma^2c^2t^2+\frac{(1-\gamma^2)^2c^2x^2}{\gamma^2v^2}+2\gamma c^2t\frac{(1-\gamma^2)x}{\gamma v} \\
\Longrightarrow \gamma^2x^2-\frac{(1-\gamma^2)^2c^2x^2}{\gamma^2v^2}+y^2+z^2 &= \gamma^2c^2t^2-\gamma^2v^2t^2+2\gamma c^2t\frac{(1-\gamma^2)x}{\gamma v}+2\gamma^2xvt \\
\Longrightarrow \left[\gamma^2-\frac{(1-\gamma^2)^2c^2}{\gamma^2v^2}\right]x^2+y^2+z^2 &= \left[\gamma^2-\gamma^2\frac{v^2}{c^2}\right]c^2t^2+\left[2c^2\frac{(1-\gamma^2)}{v}+2\gamma^2v\right]xt
\end{aligned} \tag{6}$$

Comparing with equation (4), we find that it has no xt term. Hence the coefficient of the xt term on the RHS of equation (6) must equal zero, i.e.

$$\begin{aligned}
2c^2\frac{(1-\gamma^2)}{v}+2\gamma^2v &= 0 \\
\Longrightarrow \frac{2c^2}{v}-\frac{2c^2\gamma^2}{v}+2\gamma^2v &= 0 \\
\Longrightarrow \frac{2c^2\gamma^2}{v}-2\gamma^2v &= \frac{2c^2}{v} \\
\Longrightarrow \frac{2c^2}{v}\gamma^2\left(1-\frac{v^2}{c^2}\right) &= \frac{2c^2}{v} \\
\Longrightarrow \gamma^2\left(1-\frac{v^2}{c^2}\right) &= 1 \\
\Longrightarrow \gamma &= \frac{1}{\sqrt{1-v^2/c^2}}
\end{aligned} \tag{7}$$

Using equation (7) in equations (3), we obtain the expressions for x' and t' as follows:

$$\begin{aligned}
x' &= \gamma(x-vt) \\
\Longrightarrow x' &= \frac{x-vt}{\sqrt{1-v^2/c^2}}
\end{aligned}$$

$$\begin{aligned}
t' &= \gamma t+\frac{(1-\gamma^2)x}{\gamma v} \\
&= \gamma\left[t+\frac{(1-\gamma^2)x}{\gamma^2v}\right]=\gamma\left[t+\left(\frac{1}{\gamma^2}-1\right)\frac{x}{v}\right] \\
&= \gamma\left[t+\left(1-\frac{v^2}{c^2}-1\right)\frac{x}{v}\right]=\gamma\left[t-\frac{vx}{c^2}\right] \\
\Longrightarrow t' &= \frac{t-(vx/c^2)}{\sqrt{1-v^2/c^2}}
\end{aligned}$$

We call $\gamma = \dfrac{1}{\sqrt{1-v^2/c^2}}$ the *relativistic factor* and introduce $\beta = \dfrac{v}{c}$ as the *velocity ratio*, both of which are dimensionless quantities. Then we can write

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

As per the second postulate, we must always have

$$\begin{aligned} v \leqslant c &\implies \frac{v}{c} \leqslant 1 \\ &\implies \beta \leqslant 1 \end{aligned}$$

Then we must also have

$$\begin{aligned} \beta^2 &\leqslant 1 \\ \implies 1-\beta^2 &\leqslant 1 \\ \implies \sqrt{1-\beta^2} &\leqslant 1 \\ \implies \frac{1}{\sqrt{1-\beta^2}} &\geqslant 1 \\ \implies \gamma &\geqslant 1 \end{aligned}$$

Hence, remember $\boxed{\beta \leqslant 1}$ and $\boxed{\gamma \geqslant 1}$.

Now, we have

$$\begin{aligned} x' &= \frac{x - v\,t}{\sqrt{1-v^2/c^2}} \\ &= \gamma(x - v\,t) \\ \implies x' &= \gamma\big(x - \beta\,c\,t\big) \quad [\because v = \beta\,c] \end{aligned}$$

Again, we have

$$\begin{aligned} t' &= \frac{t - (v\,x/c^2)}{\sqrt{1-v^2/c^2}} \\ &= \gamma\left(t - \frac{v\,x}{c^2}\right) \\ &= \gamma\left(t - \frac{\beta\,c\,x}{c^2}\right) = \gamma\left(t - \frac{\beta}{c}x\right) \\ \implies c\,t' &= \gamma\big(c\,t - \beta\,x\big) \end{aligned}$$

Therefore the transformation equations, from S to S', that keep the speed of light constant in both frames are as follows:

$$\boxed{\begin{aligned} x' &= \gamma\big(x - \beta\,c\,t\big) \\ y' &= y \\ z' &= z \\ c\,t' &= \gamma\big(c\,t - \beta\,x\big) \end{aligned}} \tag{8}$$

Equations (8) are known as the *Lorentz transformation equations*.

For the inverse transformation, from S' to S, we need to replace v with $-v$, which implies that β would be replaced by $-\beta$. However, the form of the equations remain the same as in equation (8) because of the first postulate.

Hence the *inverse Lorentz transformation* equations are

$$\boxed{\begin{aligned} x &= \gamma\left(x' + \beta c t'\right) \\ y &= y' \\ z &= z' \\ c t &= \gamma\left(c t' + \beta x'\right) \end{aligned}} \tag{9}$$

3.1 Upper speed limit of the universe

Suppose we assume that the relative speed between the frames is larger than the speed of light, i.e. $v > c$, in violation of the first postulate.

Then we would have

$$\begin{aligned} \frac{v}{c} > 1 &\implies \frac{v^2}{c^2} > 1 \\ &\implies 1 - \frac{v^2}{c^2} < 0 \\ &\implies \sqrt{1 - \frac{v^2}{c^2}} \text{ is imaginary} \\ &\implies \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \text{ is also imaginary} \end{aligned}$$

This implies that the space and time coordinates of events in S and S' would also be imaginary. But this is physically impossible.

Hence, our assumption is incorrect, i.e. it is not possible for the relative speed between inertial frames to be greater than the speed of light.

In other words, nothing in our universe can move faster than light!

3.2 Matrix form of the Lorentz transformation

We can re-write equation (8) as follows:

$$\begin{aligned} x' &= \gamma \cdot x + 0 \cdot y + 0 \cdot z + (-\beta\gamma) \cdot (c t) \\ y' &= 0 \cdot x + 1 \cdot y + 0 \cdot z + 0 \cdot (c t) \\ z' &= 0 \cdot x + 0 \cdot y + 1 \cdot z + 0 \cdot (c t) \\ (c t)' &= (-\beta\gamma) \cdot x + 0 \cdot y + 0 \cdot z + \gamma \cdot (c t) \end{aligned}$$

The above equations may now be collected in the form of the following matrix equation:

$$\begin{bmatrix} x' \\ y' \\ z' \\ c\,t' \end{bmatrix} = \begin{bmatrix} \gamma & 0 & 0 & -\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\beta\gamma & 0 & 0 & \gamma \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ c\,t \end{bmatrix}$$

We define the *Lorentz transformation matrix* as

$$\mathbf{L} = \begin{bmatrix} \gamma & 0 & 0 & -\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\beta\gamma & 0 & 0 & \gamma \end{bmatrix}$$

which allows us to write the *Lorentz transformation in matrix form* as follows:

$$\mathbf{x}' = \mathbf{L}\mathbf{x}$$

where $\mathbf{x}'$ and $\mathbf{x}$ are the following column matrices containing the coordinates:

$$\mathbf{x}' = \begin{bmatrix} x' \\ y' \\ z' \\ c\,t' \end{bmatrix} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \\ c\,t \end{bmatrix}$$

Likewise, the *inverse Lorentz transformation in matrix* form can be written as

$$\mathbf{x} = \mathbf{L}^{-1}\mathbf{x}'$$

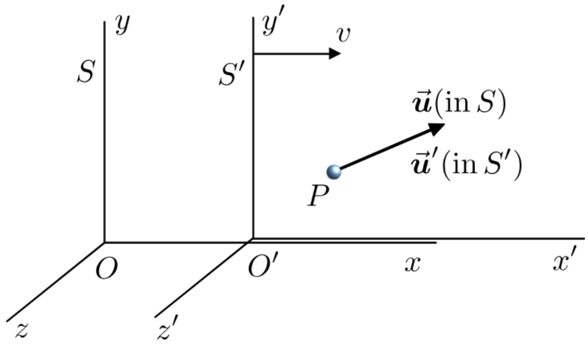
where the inverse *Lorentz transformation matrix* is

$$\mathbf{L}^{-1} = \begin{bmatrix} \gamma & 0 & 0 & \beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \beta\gamma & 0 & 0 & \gamma \end{bmatrix}$$

4 Effects of relativity

4.1 Velocity transformation

Consider two inertial frames of reference S and S' with relative velocity v along the xx'-axes. Suppose a particle at P is moving with a velocity $\mathbf{u}$ with respect to S and with a velocity $\mathbf{u}'$ with



respect to S'.

Then the components of its velocity with respect to these two frames are as follows:

$$u_x = \frac{\mathrm{d}x}{\mathrm{d}t} \quad u_y = \frac{\mathrm{d}y}{\mathrm{d}t} \quad u_z = \frac{\mathrm{d}z}{\mathrm{d}t}$$
$$u'_x = \frac{\mathrm{d}x'}{\mathrm{d}t'} \quad u'_y = \frac{\mathrm{d}y'}{\mathrm{d}t'} \quad u'_z = \frac{\mathrm{d}z'}{\mathrm{d}t'}$$

Now, from the Lorentz transformation of coordinates from S to S', we have

$$\begin{aligned} x' &= \gamma\big(x - \beta c\, t\big) \\ \implies \mathrm{d}x' &= \gamma\big(\mathrm{d}x - \beta c\, \mathrm{d}t\big) \\ \implies \mathrm{d}x' &= \gamma(\mathrm{d}x - v\, \mathrm{d}t) \end{aligned}$$

Again

$$\begin{aligned} c\, t' &= \gamma\big(c\, t - \beta x\big) \\ \implies c\, \mathrm{d}t' &= \gamma\big(c\, \mathrm{d}t - \beta\, \mathrm{d}x\big) \\ \implies \mathrm{d}t' &= \gamma\left(\mathrm{d}t - \frac{\beta}{c}\,\mathrm{d}x\right) = \gamma\left(\mathrm{d}t - \frac{v/c}{c}\,\mathrm{d}x\right) \\ \implies \mathrm{d}t' &= \gamma\left(\mathrm{d}t - \frac{v}{c^2}\,\mathrm{d}x\right) \end{aligned}$$

Finally,

$$\mathrm{d}y' = \mathrm{d}y \quad \text{and} \quad \mathrm{d}z' = \mathrm{d}z$$

Now, we obtain

$$\begin{aligned} u'_x = \frac{\mathrm{d}x'}{\mathrm{d}t'} &= \frac{\gamma(\mathrm{d}x - v\,\mathrm{d}t)}{\gamma(\mathrm{d}t - v/c^2\,\mathrm{d}x)} \\ &= \frac{\mathrm{d}t\,(\mathrm{d}x/\mathrm{d}t - v)}{\mathrm{d}t\,(1 - v/c^2 \cdot \mathrm{d}x/\mathrm{d}t)} \\ \implies u'_x &= \frac{u_x - v}{1 - u_x v/c^2} \end{aligned}$$

Likewise,

$$
\begin{aligned}
u'_y = \frac{\mathrm{d}y'}{\mathrm{d}t'} &= \frac{\mathrm{d}y}{\gamma(\mathrm{d}t - v/c^2\,\mathrm{d}x)} \\
&= \frac{\mathrm{d}y\big/\mathrm{d}t}{\gamma(\mathrm{d}t - v/c^2 \cdot \mathrm{d}x/\mathrm{d}t)} \\
\implies u'_y &= \frac{u_y\sqrt{1-\beta^2}}{1 - u_x v/c^2}
\end{aligned}
$$

Similarly,

$$
u'_z = \frac{u_z\sqrt{1-\beta^2}}{1 - u_x v/c^2}
$$

Therefore, we obtain the *Lorentz velocity transformation* equations as

$$
\boxed{
\begin{aligned}
u'_x &= \frac{u_x - v}{1 - u_x v/c^2} \\
u'_y &= \frac{u_y\sqrt{1-\beta^2}}{1 - u_x v/c^2} \\
u'_z &= \frac{u_z\sqrt{1-\beta^2}}{1 - u_x v/c^2}
\end{aligned}
}
$$

The *inverse Lorentz velocity transformation* equations would then be

$$
\boxed{
\begin{aligned}
u_x &= \frac{u'_x + v}{1 + u'_x v/c^2} \\
u_y &= \frac{u'_y\sqrt{1-\beta^2}}{1 + u'_x v/c^2} \\
u_z &= \frac{u'_z\sqrt{1-\beta^2}}{1 + u'_x v/c^2}
\end{aligned}
}
$$

Special case: Suppose the motion of P is along the xx'-axes, then we would have

$$
u_x = u \quad u_y = 0 \quad u_z = 0
$$

Then we would have $u'_x = u'$ and $u'_y = u'_z = 0$. Then we obtain the forward and inverse Lorentz velocity transformations to be

$$
\boxed{u' = \frac{u - v}{1 - uv/c^2} \quad \text{and} \quad u = \frac{u' + v}{1 + u'v/c^2}}
$$

The above equations are referred to as *Einstein's law of relativistic velocity addition.*

Example: A spaceship leaves the Earth at a speed of $0.6c$. A rocket leaves the spaceship at a speed of $0.9c$ with respect to the spaceship. Calculate the speed of the rocket with respect to the Earth if

i) it is fired in the same direction as the spaceship.

ii) it is fired in the opposite direction as the spaceship.

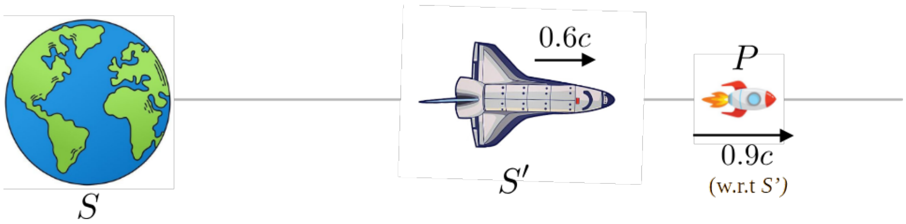
Solution: Here we consider

$$S\text{: the Earth}$$
$$S'\text{: the spaceship}$$
$$P\text{: the rocket}$$

Hence, we have

$$\text{speed of } S' \text{ w.r.t. } S = \text{speed of the spaceship w.r.t. the Earth} = |v| = 0.6c$$
$$\text{speed of } P \text{ w.r.t. } S' = \text{speed of the rocket w.r.t. the spaceship} = |u'| = 0.9c$$
$$\text{velocity of } P \text{ w.r.t. } S = \text{speed of the rocket w.r.t. the Earth} = u = ?$$

i) When the rocket is fired in the same direction as the spaceship



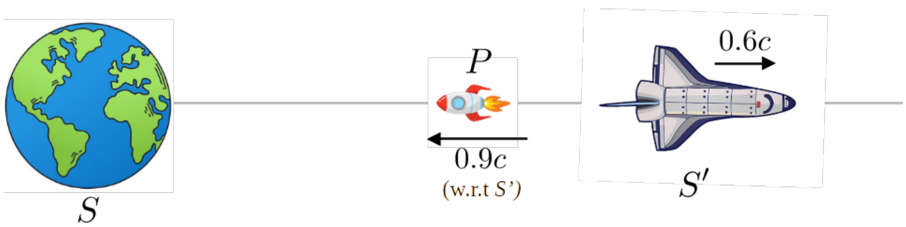
Here we have

$$v = 0.6c\,, \quad u' = 0.9c$$

Then from Einstein's law of relativistic velocity addition, we have

$$\begin{aligned} u &= \frac{u'+v}{1+u'v/c^2} \\ &= \frac{0.9c+0.6c}{1+(0.9c\times 0.6c)/c^2} \\ &= \frac{1.5c}{1+0.54c^2/c^2} \\ &= \frac{1.5c}{1.54} \\ \implies u &= 0.974c \end{aligned}$$

ii) When the rocket is fired in the opposite direction as the spaceship



Here we have

$$v = 0.6c, \quad u' = -0.9c$$

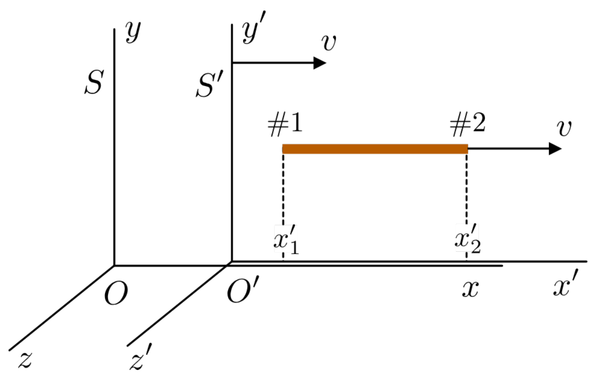
Then from Einstein's law of relativistic velocity addition, we have

$$\begin{aligned} u &= \frac{u' + v}{1 + u'v/c^2} \\ &= \frac{-0.9c + 0.6c}{1 + (-0.9c \times 0.6c)/c^2} \\ &= \frac{-0.3c}{1 - 0.54c^2/c^2} \\ &= \frac{-0.3c}{0.46} \\ \implies u &= -0.652c \end{aligned}$$

Here the $-$ sign indicates that the rocket is moving towards the Earth.

4.2 Length contraction

Suppose we have a rod of natural length L_0, whose ends are marked as #1 and #2, that is moving with a velocity v with respect to a rest frame S.



Then the rod would be at rest with respect to the moving frame S'. If x'_1 and x'_2 are the coordinates of either ends of the rod in the frame S', then the natural length of the rod is

$$L_0 = x'_2 - x'_1 \tag{1}$$

Now, if the corresponding coordinates of either end of the rod in the rest frame S are x_1 and x_2 respectively, then the length of the rod, as measured in the frame S is

$$L = x_2 - x_1 \tag{2}$$

Using Lorentz transformation, if we measure the coordinates x_1 and x_2 simultaneously, we have

$$\begin{aligned} x'_1 = \gamma\big(x_1 - \beta c t\big) \quad &\text{and} \quad x'_2 = \gamma\big(x_2 - \beta c t\big) \\ \implies x'_1 = \gamma(x_1 - v t) \quad &\text{and} \quad x'_2 = \gamma(x_2 - v t) \end{aligned} \tag{3}$$

Inserting equations (3) in equation (1), we have

$$\begin{aligned}
L_0 &= \gamma(x_2 - vt) - \gamma(x_1 - vt) \\
&= \gamma[x_2 - vt - x_1 + vt] \\
&= \gamma(x_2 - x_1) \\
\implies L_0 &= \gamma L \quad \text{[using eqn. (2)]}
\end{aligned}$$

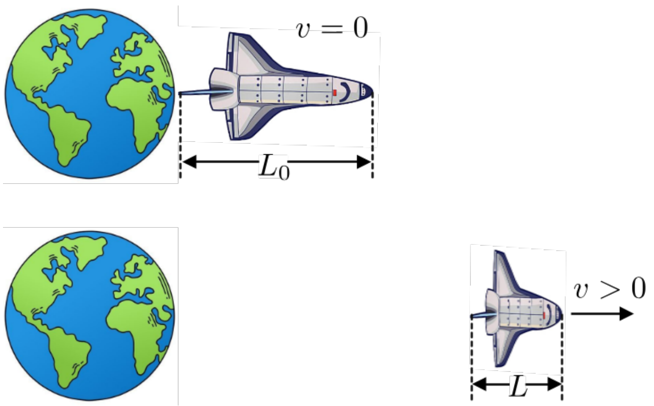
$$\implies \boxed{L_0 = \frac{L}{\sqrt{1-\beta^2}}} \quad \text{or} \quad \boxed{L = L_0\sqrt{1-\beta^2}} \tag{4}$$

We know that

$$\begin{aligned}
\gamma &\geqslant 1 \\
\implies \frac{L_0}{L} &\geqslant 1 \\
\implies L &\leqslant L_0
\end{aligned}$$

Therefore, to a stationary observer in the frame S, the rod seems as if it has contracted parallel to the direction of its motion.

This phenomenon of length contraction is known as the **Lorentz-Fitzgerald contraction**.



The faster something moves, the more squashed it appears to someone observing from a stationary frame. This happens the other way around as well.